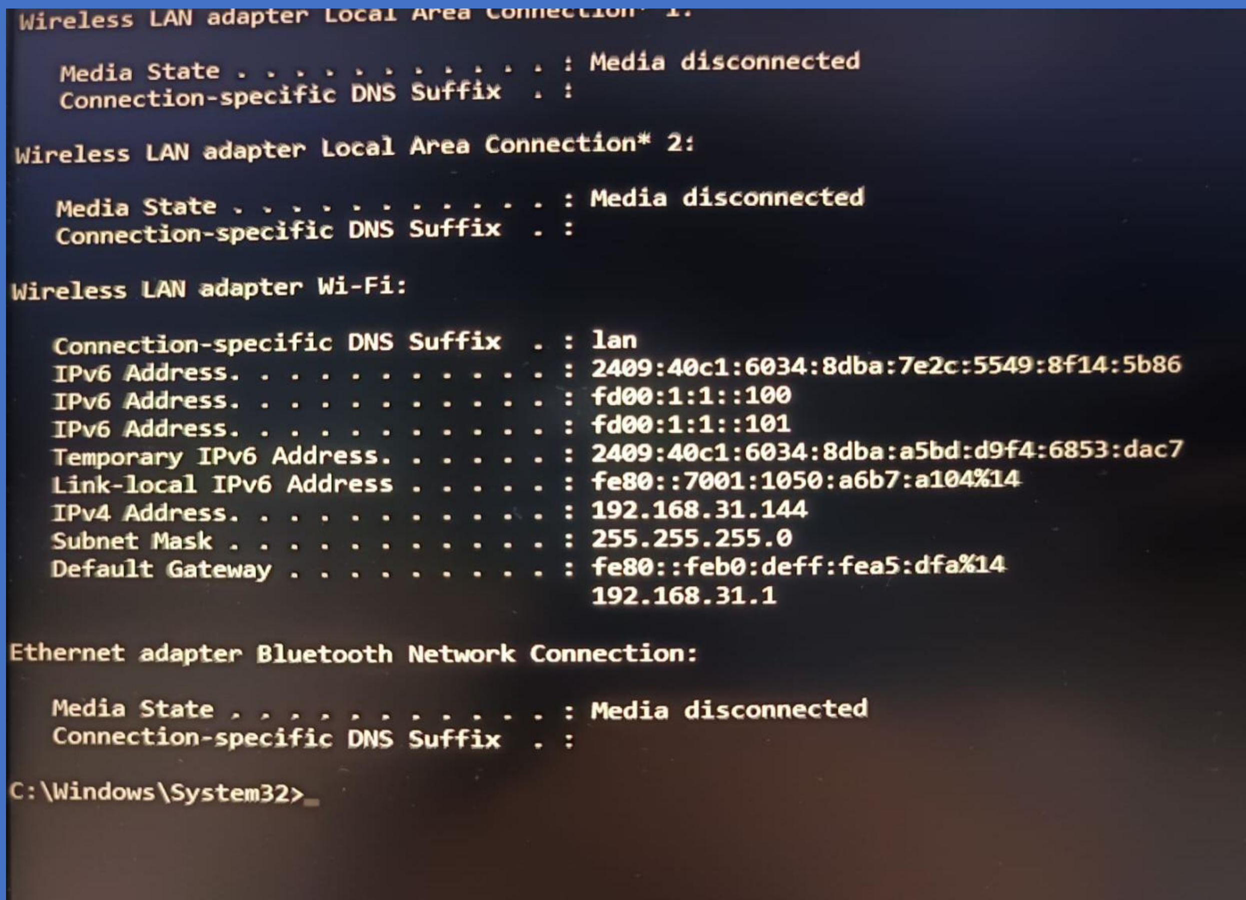
Task

1. Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.



```
Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : lan
   IPv6 Address. . . . . . . . . . . : 2409:40c1:6034:8dba:7e2c:5549:8f14:5b86
   IPv6 Address. . . . . . . . . . . : fd00:1:1::100
   IPv6 Address. . . . . . . . . . . : fd00:1:1::101
   Temporary IPv6 Address. . . . . . : 2409:40c1:6034:8dba:a5bd:d9f4:6853:dac7
   Link-local IPv6 Address . . . . . : fe80::7001:1050:a6b7:a104%14
   IPv4 Address. . . . . . . . . . . : 192.168.31.144
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::feb0:deff:fea5:dfa%14
                                       192.168.31.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

C:\Windows\System32>_
```

2. Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

C:\Windows\System32>ping www.google.com

Pinging www.google.com [2001:4860:4826:7700::] with 32 bytes of data:

Reply from 2001:4860:4826:7700::: time=51ms

Reply from 2001:4860:4826:7700::: time=51ms

Reply from 2001:4860:4826:7700::: time=39ms Reply

from 2001:4860:4826:7700::: time=47ms

Ping statistics for 2001:4860:4826:7700:::

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate

round trip times in milli-seconds:

Minimum = 39ms, Maximum = 51ms, Average = 47ms

3. Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

```
C:\Windows\System32>tracert www.flipkart.com

Tracing route to flipkart.com [163.53.76.86] over
a maximum of 30 hops:

1       145 ms     5 ms     1 ms  jiofiber.local.html [192.168.31.1]
2       5 ms     2 ms     1 ms  192.0.0.1
3       38 ms    37 ms    33 ms  192.0.0.1
4       27 ms    34 ms    18 ms  192.0.0.1
5       32 ms    28 ms    25 ms  192.0.0.1
6       23 ms    18 ms    23 ms  192.168.227.195
7       32 ms    26 ms    33 ms  192.168.188.62
8       *        *        *     Request timed out.
9       *        *        *     Request timed out.
10      65 ms    50 ms    49 ms  115.242.187.186.static.jio.com [115.242.187.186]
11      *        *        *     Request timed out.
12      *        *        *     Request timed out.
13      *        *        *     Request timed out.
14      *        *        *     Request timed out.
15      43 ms    47 ms    50 ms  163.53.76.86

Trace complete.
```

4. Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

```
C:\Windows\System32>nslookup www.spotify.com
```

Server: jiofiber.local.html

Address: 192.168.31.1

Non-authoritative answer:

Name: atc.spotify.map.fastly.net

Addresses: 2a04:4e42:31::810

151.101.211.42

Aliases: www.spotify.com

5. A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

Hint: Think about checking both network connectivity and DNS resolution.

Command	Purpose
ipconfig	Checks the computer's IP address, subnet mask, and default gateway.
ping www.zomato.com	Checks whether the computer can reach the Zomato server over the network.
nslookup www.zomato.com	Checks whether the DNS server can resolve the website name to an IP address.